

#LP: Find the zeroes of each of the following quadratic polynomial and verify the relationship between the zeroes and their coefficients:

(i) $g(s) = 4s^2 - 4s + 1$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \cdot \beta = \frac{c}{a}$$

$$(ii) q(x) = \sqrt{3x^2 + 10x + 7\sqrt{3}} = 0 \quad p+q=10$$

$$\Rightarrow \sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0 \quad pq = 7\sqrt{3} \times \sqrt{3} = 21$$

$$\rightarrow \sqrt{3}x^2 + \sqrt{3}\sqrt{3}x + 7x + 7\sqrt{3} = 0$$

$$\rightarrow \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$$

$$\Rightarrow (x + \sqrt{3})(\sqrt{3}x + 7) = 0$$

$$x + \sqrt{3} = 0 \quad | \quad \sqrt{3}x + 7 = 0$$

$$x = -\sqrt{3}$$

$$x = -\frac{7}{\sqrt{3}}$$

$$(iii) f(v) = v^2 + 4\sqrt{3}v - 15$$

$$\sqrt{3}x^2 + 10x + 7\sqrt{3}$$

$$\alpha = -\sqrt{3} \quad \beta = -\frac{7}{\sqrt{3}}$$

$$\alpha + \beta = -\frac{b}{a}$$

$$-\sqrt{3} + \left(-\frac{7}{\sqrt{3}}\right) = -\frac{10}{\sqrt{3}}$$

$$\rightarrow \frac{-3-7}{\sqrt{3}} = -\frac{10}{\sqrt{3}}$$

$$\Rightarrow \frac{-10}{\sqrt{3}} = -\frac{10}{\sqrt{3}}$$

$$LHS = RHS$$

$$\alpha \cdot \beta = \frac{c}{a}$$

$$-\sqrt{3} \times -\frac{7}{\sqrt{3}} = \frac{7\sqrt{3}}{\sqrt{3}}$$

$$7 = 7$$

$$LHS = RHS$$

H.P.

$$(ii) q(x) = \sqrt{3x^2 + 10x + 7\sqrt{3}} = 0 \quad p+q=10$$

$$\Rightarrow \sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0 \quad pq = 7\sqrt{3} \times \sqrt{3} = 21$$

$$\sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$$

$$(x + \sqrt{3})[\sqrt{3}x + 7] = 0$$

$$x + \sqrt{3} = 0$$

$$x = -\sqrt{3}$$

$$\sqrt{3}x + 7 = 0$$

$$x = -\frac{7}{\sqrt{3}}$$

$$(iii) f(v) = v^2 + 4\sqrt{3}v - 15$$

$$\sqrt{3}x^2 + 10x + 7\sqrt{3}$$

$$\begin{matrix} -\sqrt{3} & \alpha \\ -\frac{7}{\sqrt{3}} & \beta \end{matrix}$$

$$\alpha + \beta = -\frac{b}{a}$$

$$-\frac{\sqrt{3}}{1} + \left(-\frac{7}{\sqrt{3}}\right) = -\frac{(10)}{\sqrt{3}}$$

$$\Rightarrow \frac{-3-7}{\sqrt{3}} = -\frac{10}{\sqrt{3}}$$

$$\Rightarrow \frac{-10}{\sqrt{3}} = -\frac{10}{\sqrt{3}}$$

$$LHS = RHS$$

$$\alpha \cdot \beta = \frac{c}{a}$$

$$-\sqrt{3} \times -\frac{7}{\sqrt{3}} = \frac{7\sqrt{3}}{\sqrt{3}}$$

$$7 = 7$$

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$$(ii) q(x) = \sqrt{3x^2 + 10x + 7\sqrt{3}} = 0 \quad p+q=10$$

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$$\rightarrow \sqrt{3}x^2 + \sqrt{3}\sqrt{3}x + 7x + 7\sqrt{3} = 0$$

$$\rightarrow \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$$

$$\Rightarrow (x + \sqrt{3})(\sqrt{3}x + 7) = 0$$

$$x + \sqrt{3} = 0 \quad \bigg| \quad \sqrt{3}x + 7 = 0$$

$$x = -\sqrt{3}$$

$$x = -\frac{7}{\sqrt{3}}$$

$$(iii) f(v) = v^2 + 4\sqrt{3}v - 15$$

soo
(11)

$$\boxed{v^2 + 4\sqrt{3}v - 15} = 0$$

$$\Rightarrow \underline{v^2 + 5\sqrt{3}v - \sqrt{3}v - 15} = 0$$

$$\Rightarrow \underline{v(v + 5\sqrt{3})} = \underline{\sqrt{3}(v + 5\sqrt{3})} = 0$$

$$\Rightarrow (v + 5\sqrt{3})(v - \sqrt{3}) = 0$$

$$v + 5\sqrt{3} = 0$$

$$\boxed{v = -5\sqrt{3}}$$

$$v - \sqrt{3} = 0$$

$$\boxed{v = \sqrt{3}}$$

$$\boxed{v^2 + 4\sqrt{3}v - 15} \begin{cases} -5\sqrt{3} \dots \alpha \\ \sqrt{3} \dots \beta \end{cases}$$

$$\boxed{\alpha + \beta = -\frac{b}{a}}$$

$$-5\sqrt{3} + \sqrt{3} = -\frac{(4\sqrt{3})}{1}$$

$$-4\sqrt{3} = -4\sqrt{3}$$

$$\boxed{\text{LHS} = \text{RHS}}$$

HP

$$\boxed{\alpha \cdot \beta = \frac{c}{a}}$$

$$-5\sqrt{3} \times \sqrt{3} = -15$$

$$-15 = -15$$

$$\boxed{\text{LHS} = \text{RHS}}$$

~~HP~~

#K3B

~~$x^2 + x - 6$~~ $\rightarrow$ 3 terms

$\Rightarrow x^2 + 3x - 2x - 6 \rightarrow$ 4 terms

$\Rightarrow \underline{x(x+3)} = \underline{-2(x+3)}$

$\Rightarrow (x+3)(x-2)$

common लिखा आरंभ है

(v) $f(x) = x^2 - (\sqrt{3} + 1)x + \sqrt{3}$

$$\boxed{x^2 - \sqrt{3}x - x + \sqrt{3}}$$

4 terms

$\Rightarrow$ $\nexists$ directly
common ~~term~~

$$- \underbrace{x(x - \sqrt{3})} = - \underbrace{1(x - \sqrt{3})} = 0$$

$$(x - \sqrt{3}) / (x - 1) = 0$$

$$x - \sqrt{3} = 0$$

$$x = \sqrt{3}$$

$$x - 1 = 0$$

$$x = 1$$

#LP: Find the zeroes of the quadratic polynomial $2x^2 - (1 + 2\sqrt{2})x + \sqrt{2}$ and verify the relationship between the zeroes and coefficients of the polynomial.

$$2x^2 - (1 + 2\sqrt{2})x + \sqrt{2} = 0$$

$$\underbrace{2x^2 - x}_{\text{1st 2 terms}} - \underbrace{2\sqrt{2}x + \sqrt{2}}_{\text{last 2 terms}} = 0$$

→ 4 terms

$$\underbrace{x(2x - 1)}_{\text{1st 2 terms}} - \underbrace{\sqrt{2}(2x - 1)}_{\text{last 2 terms}} = 0$$

$$(2x - 1)(x - \sqrt{2}) = 0$$

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

$$x - \sqrt{2} = 0$$

$$x = \sqrt{2}$$

Verify

#LP: If α and β are the zeroes of the polynomial $5x^2 + 3x - 7$, find the value of:

1. $\alpha + \beta$

$$\Rightarrow -\frac{3}{5} \checkmark$$

$$5x^2 + 3x - 7$$

α
 β

$$\begin{aligned} \alpha + \beta &= -\frac{3}{5} \\ \alpha \cdot \beta &= -\frac{7}{5} \end{aligned} \quad \text{--- (I)}$$

2. $\alpha^2 + \beta^2$

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\Rightarrow (\alpha + \beta)^2 - 2\alpha\beta = \alpha^2 + \beta^2$$

$$\begin{aligned} \Rightarrow \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= \left(-\frac{3}{5}\right)^2 - 2 \times \left(-\frac{7}{5}\right) \\ &= \frac{9}{25} + \frac{14}{5} \Rightarrow \left(\checkmark\right) \end{aligned}$$

$$3. \frac{1}{\alpha} + \frac{1}{\beta}$$

$$\Rightarrow \frac{1(\beta) + 1(\alpha)}{\alpha\beta}$$

$$\Rightarrow \frac{\alpha + \beta}{\alpha\beta} \Rightarrow \frac{+3}{\cancel{4} \frac{7}{\cancel{8}}} \Rightarrow \boxed{\frac{3}{7}}$$

$$4. \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$$

$$\Rightarrow \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$\Rightarrow \frac{\left(\frac{-7}{2}\right)^2}{\frac{-7}{5}} \quad \checkmark$$

$$\left(\alpha + \frac{1}{\beta}\right) \times \left(\beta + \frac{1}{\alpha}\right) \Rightarrow$$

$$\rightarrow \alpha\beta + \cancel{\alpha} \cdot \frac{1}{\cancel{\beta}} + \frac{1}{\cancel{\beta}} \cdot \beta + \frac{1}{\beta} \times \frac{1}{\alpha}$$

$$\rightarrow \alpha\beta + 1 + 1 + \frac{1}{\cancel{\alpha}\beta}$$

$$\rightarrow \alpha\beta + 2 + \frac{1}{\cancel{\alpha}\beta}$$

$$\rightarrow \left(-\frac{7}{5}\right) + 2 + \frac{1}{\left(-\frac{7}{5}\right)} \checkmark$$

$$(1+p)(p+q)$$

$$p+q+p^2+pq$$

Tw

$$\alpha^2\beta + \alpha\beta^2$$

Tw

$$\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta$$

#LP: 

If ' α ' and ' β ' are the zeroes of the polynomial $p(y) = y^2 - 5y + 3$, then find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. (2025)